Mock Interview Edition

Harmful Content Detection: Rare Classes, Labels and a Moderation Cascade

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every scenario, by part

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02 Price the rarity Estimation 6

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

Round 1 Framing, rarity and metrics 4

01 Why eight labels and not one? Taxonomy 4
02 Price the rarity Estimation 6
03 Why not accuracy, and why not ROC? Metric 7
04 Where does the classifier end? Scope 8
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11 When does domain adversarial training pay? Domain shift 19
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17 What does the paper say its model gets wrong? Limits 28

SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: taxonomy, rarity, metric and the line with policy

Defining what counts as undesired, pricing the rarity of the cases that matter, choosing a metric that survives class imbalance, and deciding where the classifier stops.

01 Why eight labels and not one?

TAXONOMY

SCENARIO

A product manager wants a single score called undesired. The paper has five categories and subcategories, and its model predicts 8 labels. Argue for the taxonomy and say what it costs.

OPENING PROBE

What does the model predict?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What would one bucket lose?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What do subcategories do for labeling?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does context do to the label?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 Price the rarity

ESTIMATION

SCENARIO

The paper says that among sampled user prompts only 0.04% included self-harm and 0.017% included hateful content with threats. Your team will label 2,000 items a round. Plan the data collection, using the paper's own Table 4 rates for random and active selection.

OPENING PROBE

What does random sampling give you?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does active selection give?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How many labels for 100 positives?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Is rare the same as negligible?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Why not accuracy, and why not ROC?

METRIC

SCENARIO

A candidate says: our self-harm model has 99.9% accuracy and a ROC curve that reaches 80% true positive rate at a 1% false positive rate, so it is excellent. The paper reports AUPRC. Show what the numbers hide.

OPENING PROBE

What does accuracy hide?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does that ROC point mean?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the precision?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does AUPRC compare against?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Where does the classifier end?

SCOPE

SCENARIO

Leadership asks the team to ship content moderation and expects the model to decide what is removed. The paper builds a classifier and says it should be paired with fair and robust enforcement practices. Draw the line and show why it helps.

OPENING PROBE

What does the paper build?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why separate them?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a threshold cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What goes in the policy layer?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Data and labels: instructions, audits and active learning

Writing labeling instructions people can follow the same way, auditing when positives are rare, and spending a labeling budget on the examples that teach the most.

05 Make the instructions followable

LABELING

SCENARIO

Annotators disagree on borderline cases, and the model trained on their labels is erratic. The paper says that detailed definitions, mutually exclusive categories and calibration sessions help. It also discusses disagreement in a more nuanced way. Describe the process and its limits.

OPENING PROBE

What are the steps?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why does this matter for the model?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the paper say about disagreement?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you do with both?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 How do you audit a rare class?

QUALITY

SCENARIO

An auditor proposes checking 1,000 randomly chosen labeled items per week and reporting the agreement rate. The paper does something different. Explain why the proposal fails for self-harm, and describe the paper's audit.

OPENING PROBE

What does a random audit see?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does agreement rate measure here?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the paper audit?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does this audit not give?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Mix the selection pipelines

ACTIVE LEARNING

SCENARIO

Your first model flags few self-harm items. A colleague proposes labeling only the items with the highest scores each round, since those are most likely to be positive. The paper uses three pipelines in parallel. Explain why.

OPENING PROBE

What does a top-score-only set miss?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What are the paper's three pipelines?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the random arm protect?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the experiment show?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the selection function

CODE REVIEW

SCENARIO

A colleague writes the selection step below for each labeling round and says it implements active learning. After three rounds, recall measured on a random test set is worse for a pattern of phrasing the model scored low, and nobody can estimate production prevalence. Find the faults.

```
def select_for_labeling(scored, budget=2000):
    # scored: list of (item, score) for one category, from the latest model
    ranked = sorted(scored, key=lambda pair: -pair[1])
    return [item for item, _ in ranked[:budget]]
```

OPENING PROBE

What is the first fault?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the second?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the third?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The model: heads, synthetic data, domain adaptation and probing

Why one head per label, what synthetic data helps and hurts, when a domain adaptation trick pays, and how to find out what a model is really looking at.

09 Eight heads on one backbone

ARCHITECTURE

SCENARIO

The paper's model is a lightweight transformer decoder initialized from a GPT model, with the output layer replaced by 8 separate MLP heads. A reviewer asks why not one output layer of 8 numbers, and whether the paper's claim of fewer parameters holds.

OPENING PROBE

What is the design?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why separate heads?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Count the parameters.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

So is the claim wrong?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

10 Patch weak spots with synthetic data

DATA

SCENARIO

A rare category has almost no public data, and an early model gives a high hate score to a neutral phrase with an identity word. The paper uses synthetic data in three ways and also found a limit. Plan it.

OPENING PROBE

What is the cold-start use?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the rare-category use?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the counterfactual use?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What was the limit?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

11 When does domain adversarial training pay?

DOMAIN SHIFT

SCENARIO

Models trained on public data do poorly on production prompts, which are longer and often contain few-shot prompts. The paper adds domain adversarial training. A teammate says it gave a 346% gain in self-harm and wants it in every model. Read Table 5 before you answer.

OPENING PROBE

What is the method and its setup?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does the table say about the mean?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What about individual rows?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What does the paper suspect?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you decide?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

12 What is the model really looking at?

PROBING

SCENARIO

The model gives a high hate score to the phrase all immigrants. A researcher says the model has learned hate. The paper investigates exactly this. Work out what is happening and how to detect it at scale.

OPENING PROBE

What does the data say about the phrase?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How does key token probing work?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did it find?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What did red-teaming add?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

13 Review the key token finder

CODE REVIEW

SCENARIO

A colleague implements input reduction to find what the model keys off. The tokens it returns never trigger the model when you feed them back, and the job takes days on a sample of a few thousand texts. The paper removes tokens greedily until the prediction drops below 0.8. Find the faults.

```
def key_tokens(tokens, score_fn, threshold=0.8):
    kept = list(tokens)
    while len(kept) > 1:
        best = max(range(len(kept)), key=lambda i: score_fn(kept[:i] + kept[i+1:]))
        kept = kept[:best] + kept[best+1:]
        if score_fn(kept) < threshold:
            break
    return kept
```

OPENING PROBE

What does it return?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What other fault is there?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

PART 4

Serving, streaming, evaluation and operations

Moderating a stream with a cheap filter and an expensive judge, reading a comparison table honestly, and running the system responsibly.

14 Price the streaming cascade

DESIGN

SCENARIO

The paper scores finished text. You must moderate an LLM's response as it streams, using the paper's classifier as tier 1 and an LLM judge as tier 2 for uncertain cases. The paper describes none of this. Size the design before you propose it.

OPENING PROBE

What does the judge cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does tier 1 cost?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does the whole cascade cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What is the trade-off in checking less often?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What is the cascade's ceiling?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

15 Review the stream gate

CODE REVIEW

SCENARIO

A colleague writes the stream gate below. In testing, flagged text still appears on screen before it is removed, some escalations seem to do nothing, and long responses get slow. Find the faults.

```
def moderate_stream(chunks, score_fn, judge_fn, low=0.3, high=0.7):
    text = ""
    for chunk in chunks:
        text += chunk
        score = score_fn(text)
        yield chunk                      # release immediately
        if score > high:
            yield "[removed]"
            return
        if score > low:
            judge_fn(text)               # escalate
```

OPENING PROBE

When is text released?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What does the judge call do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does each check cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What is missing at the end?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix, and a test.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

16 Read the comparison table honestly

EVALUATION

SCENARIO

A vendor slide says: our model beats Perspective API everywhere. The paper's Table 3 compares the two. Say what the table supports and what it does not.

OPENING PROBE

Where does the paper's model win?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Where does Perspective API win?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the stars mean?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

17 What does the paper say its model gets wrong?

LIMITS

SCENARIO

A launch review asks for the known limits of the classifier. The paper has a section on them. List them and say how you would monitor each one.

OPENING PROBE

What are the stated limits?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What about bias?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How would you monitor?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you not claim?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

18 Data consent, annotator welfare and ten times the traffic

OPERATIONS

SCENARIO

Your company wants to train a moderation model on customer traffic and expects ten times the volume next year. List the obligations and what scales.

OPENING PROBE

What does the paper say about data use?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does it say about annotators?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What scales with ten times the traffic?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does not scale?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 26

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Why eight labels and not one?	6			
02	Price the rarity	5			
03	Why not accuracy, and why not ROC?	5			
04	Where does the classifier end?	5			
05	Make the instructions followable	5			
06	How do you audit a rare class?	5			
07	Mix the selection pipelines	5			
08	Review the selection function	5			
09	Eight heads on one backbone	5			
10	Patch weak spots with synthetic data	5			
11	When does domain adversarial training pay?	6			
12	What is the model really looking at?	5			
13	Review the key token finder	5			
14	Price the streaming cascade	6			
15	Review the stream gate	6			
16	Read the comparison table honestly	5			
17	What does the paper say its model gets wrong?	5			
18	Data consent, annotator welfare and ten times the...	5			